QPT status: runs finished; recovery gate not met

Checked 6 October 2026, 15:32 CEST

What is complete

The 352-pair calibration is complete and audited. The separate 24-pair validation array 2014377 finished on 5 October at 22:08 CEST. All 24 JSON/NPZ archives, checksums, frozen settings and source provenance now verify locally. No QPT jobs are currently queued or running.

This is the corrected n = 2, full-128-row, noiseless rank-one comparison: 8 new channels, one initialization per channel and 3 FRAMES sampling streams. Validation used 10,000 row epochs (1.28 million nominal row accesses per method when completed), five times the calibration budget. These are not n = 4 or high-qubit results.

Final validation outcomes

Check	Quiroga (printed adaFGD)	FRAMES
Finished normally	5 / 8	24 / 24
Numerical failures	3 / 8	0 / 24
Fidelity ≥ 0.99	0 / 8 pass	24 / 24 pass
Normalized TP error ≤ 0.001	5 / 8 pass	19 / 24 pass
Full-design RMSE ≤ 1e-6	0 / 8 pass	0 / 24 pass
Full recovery gate (all three)	0 / 8 pass	0 / 24 pass

Quiroga repetitions were verified identical and counted once per channel; 24 FRAMES runs are nested in 8 independent channels. The 9 failed paired archives represent 3 Quiroga channel failures, not 9 independent failures. All failures are retained; no result is missing.

What this means

FRAMES is numerically stable in these runs and has very high fidelity, but measurement fit remains the common blocker: even its best RMSE is 7.25e-6, above 1e-6. Its median across channel means is 2.88e-5; five runs also fail TP. High fidelity alone is not successful recovery.

Quiroga has two separate problems here: numerical divergence on channels 100, 102 and 106, and poor reconstruction on the five channels that finish. Those five pass TP, but fidelity ranges only from 0.0216 to 0.2207. This is not merely a TP-gate issue.

Where the project stands / next decision

Execution and result preservation are working; the benchmark is not yet ready for a successful-recovery claim. The frozen choices were independently tuned: Quiroga fixed alpha = 0.5, step multiplier 0.01; FRAMES initial alpha = 5e-5, increasing with exponent 0.25. Both penalties landed at grid boundaries, and neither selection passed the calibration gate.

Recommended next step: diagnose Quiroga stability and reconstruction, then refine tuning on calibration data before extending this comparison to larger qubit counts. Keep this validation result fixed; do not rerank on it. Any revised method/configuration needs fresh confirmation. No new jobs or parameter changes were made for this status check; the final held-out split remains untouched.

Source c7a8c9789b78 • selection fa9b5d5287f26029 • independent aggregation cross-check agrees

Evidence: /home/tony/qpt_quiroga_status_20261006_1529/validation_audit.json